

Paper 4C: Further Mechanics 2

Topics	What students need to learn:		
	Content		Guidance
1 Motion in a circle	1.1	Angular speed. $v = r\omega$. Uniform motion of a particle moving in a horizontal circle. Radial acceleration in circular motion. The forms $r\omega^2$ and $\frac{v^2}{r}$ are required.	Problems involving the 'conical pendulum', an elastic string, motion on a banked surface, as well as other contexts, may be set.
	1.2	Motion of a particle in a vertical circle. Radial and tangential acceleration in circular motion. Kinetic and potential energy and the conservation of energy principle applied to motion in a vertical circle.	Questions may be set which involve complete or incomplete circles.